

Started on Friday, 22 May 2026, 8:59 PM**State** Finished**Completed on** Friday, 22 May 2026, 9:05 PM**Time taken** 5 mins 25 secs**Marks** 3.00/3.00**Grade** 10.00 out of 10.00 (100%)Question **1**

Correct

Mark 1.00 out of 1.00

Write a python program to find the 1-Norm of a matrix and display the results in two decimal places.

For example:

Input	Result
<code>[[-1, 3],[3, -4],[1, 7]]</code>	14.00

Answer: (penalty regime: 0 %)

```

1  '''
2  Program to find 1-norm of a matrix.
3  Developed by: THENAMUDHEN S
4  RegisterNumber: 212225040471
5  '''
6  import os
7  os.environ["OPENBLAS_NUM_THREADS"]="1"
8  import numpy as np
9  matrix=eval(input())
10 one_matrix=np.linalg.norm(matrix,1)
11 print("{:.2f}".format(one_matrix))
12

```

	Input	Expected	Got	
✓	<code>[[-1, 3],[3, -4],[1, 7]]</code>	14.00	14.00	✓
✓	<code>[[1, 2, 3],[-3,-4,-1],[9,6,1]]</code>	13.00	13.00	✓

Passed all tests! ✓

► Show/hide question author's solution (Python3)

Correct

Marks for this submission: 1.00/1.00.

Question **2**

Correct

Mark 1.00 out of 1.00

Write a program to find L2-norm of a matrix and display the result in two decimal places.

For example:

Input	Result
[[1,2],[3,4]]	5.46
[[-1, 3],[3, -4],[1, 7]]	8.66

Answer: (penalty regime: 0. %)

Reset answer

```

1  '''
2  Program to find 2-norm of a matrix.
3  Developed by: THENAMUDHEN S
4  RegisterNumber: 212225040471
5  '''
6  import os
7  os.environ["OPENBLAS_NUM_THREADS"]="1"
8  import numpy as np
9  matrix=eval(input())
10 two_matrix=np.linalg.norm(matrix,2)
11 print("{:.2f}".format(two_matrix))
12

```

	Input	Expected	Got	
✓	[[1,2],[3,4]]	5.46	5.46	✓
✓	[[-1, 3],[3, -4],[1, 7]]	8.66	8.66	✓
✓	[[2, 3],[3, 4],[1, 8]]	9.86	9.86	✓

Passed all tests! ✓

► Show/hide question author's solution (Python3)

Correct

Marks for this submission: 1.00/1.00.

Question **3**

Correct

Mark 1.00 out of 1.00

Write a program to find the Infinity of a matrix and display the result in two decimal places.

For example:

Input	Result
<code>[[-1, 3],[3, -4],[1, 7]]</code>	8.00

Answer: (penalty regime: 0. %)

```

1  '''
2  Program to find infinity-norm of a matrix.
3  Developed by: THENAMUDHEN S
4  RegisterNumber: 212225040471
5  '''
6  import os
7  os.environ["OPENBLAS_NUM_THREADS"]="1"
8  import numpy as np
9  matrix=eval(input())
10 inf_matrix=np.linalg.norm(matrix,np.inf)
11 print("{:.2f}".format(inf_matrix))

```

	Input	Expected	Got	
✓	<code>[[-1, 3],[3, -4],[1, 7]]</code>	8.00	8.00	✓
✓	<code>[[1,2,3],[-9,-8,-3],[10,3,2]]</code>	20.00	20.00	✓

Passed all tests! ✓

► Show/hide question author's solution (Python3)

Correct

Marks for this submission: 1.00/1.00.